Wednesday LAB Questions (11-sept-2024)

1. Write a program that prints the sum of the first n Even numbers using a function.
   Example:
   ```
   Input: Enter the value of n: 5
   Output: Sum of first 5 even numbers: 30
   ```

2. Write a program to check if a given number is an Armstrong number or not using a function. An Armstrong number is a number that is equal to the sum of the cubes of its digits.
   Example:
   ```
   Input: Enter a number: 153
   Output: 153 is an Armstrong number
   ```

3. Write a program to convert a decimal number to its binary equivalent using functions.
   Example:
   ```
   Input: Enter a decimal number: 10
   Output: Binary equivalent: 1010
   ```

4. Write a program that checks whether a number is a "strong number". A strong number is a number whose sum of the factorial of digits is equal to the original number. Use multiple functions: one to check if a number is strong, one to calculate the factorial of a digit, and one for input and output.
   Example:
   ```
   Input: Enter a number: 145
   Output: 145 is a strong number (1! + 4! + 5! = 145)
   ```

5. Write a program to check if a number is a Harshad (Niven) number. A number is called a Harshad number if it is divisible by the sum of its digits. Use multiple functions to check divisibility and calculate the sum of digits.
   Example:
   ```
   Input: Enter a number: 18
   Output: 18 is a Harshad number (18 / (1+8) = 2)
   ```